

AquaSmart Analytical Report

RISK: HIGH

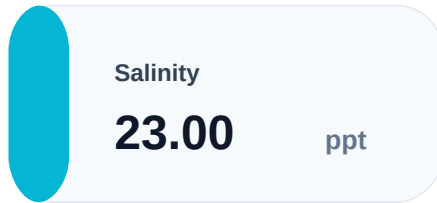
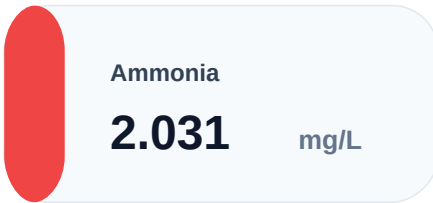
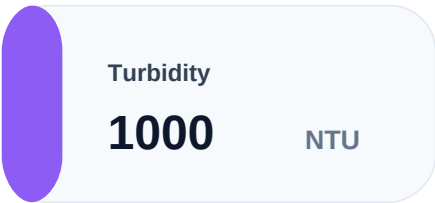
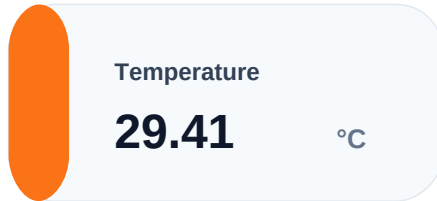
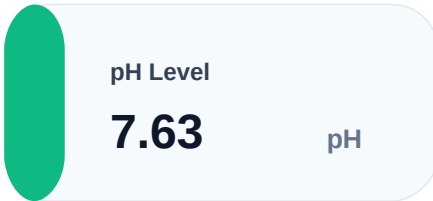
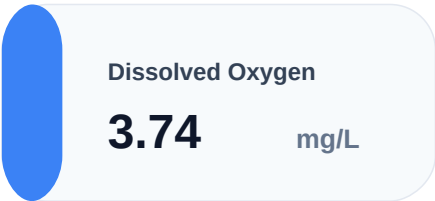
Report ID: AS-597038

Pond: Research Pond A | Pond ID: 1

Latest Reading: 2/22/2026, 11:41:41 PM

Generated: 2/26/2026, 6:59:57 PM

Researcher: Research Group 16



Context Snapshot

Location: Colombo, Sri Lanka

Weather: Clear Sky | Temp: 28°C | Humidity: 84%

Wind: 18 km/h | Rain Chance: 53%

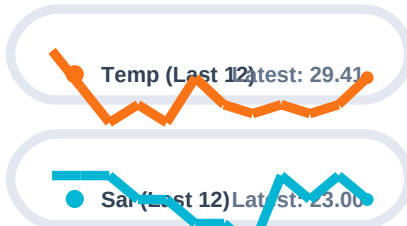
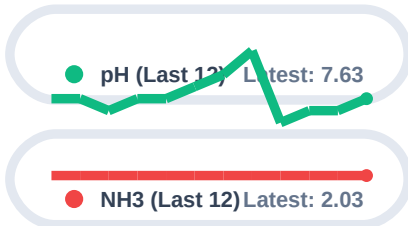
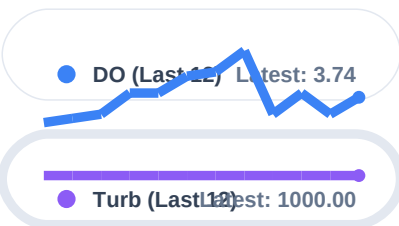
AI Water Quality Prediction

Next: Forecast not available (enable 'Show AI Forecast' to generate).

COLLECTING DATA...

Recent Trends & Quality Stats

Last 12 readings summary (sparklines) + distribution



Quality Distribution (last samples)

OK: 0 | Warning: 0 | Critical: 12

1) Pond Sensor Matrix (Real-time)

Parameter	Current (with time)	Status
Dissolved Oxygen	3.74 mg/L @ 11:41 PM	CRITICAL
pH Level	7.63 pH @ 11:41 PM	OPTIMAL
Temperature	29.41 °C @ 11:41 PM	STABLE
Turbidity	1000 NTU @ 11:41 PM	CRITICAL
Ammonia (NH3)	2.031 mg/L @ 11:41 PM	TOXIC
Salinity	23.00 ppt @ 11:41 PM	NORMAL

2) System Control Modes

Main Aerator

ON

AUTO

OFF

Water Pump

ON

AUTO

OFF

pH Doser

ON

AUTO

OFF

Feed Pump

ON

AUTO

OFF

Temp Control

ON

AUTO

OFF

Bio-Filter

ON

AUTO

OFF

3) Alerts Summary

Critical: 2

Warnings: 0

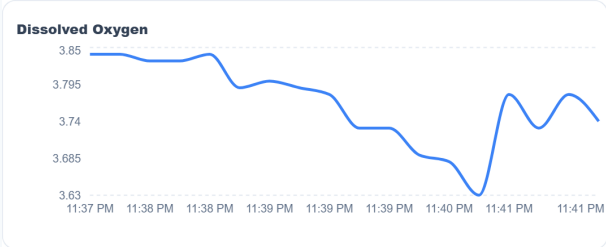
2 NEW

- 06:59 PM — Critical Oxygen Level: DO dropped to 3.74 mg/L. Activating aerators recommended.
- 06:59 PM — Toxic Ammonia: Ammonia at 2.031 mg/L. Immediate water exchange required.

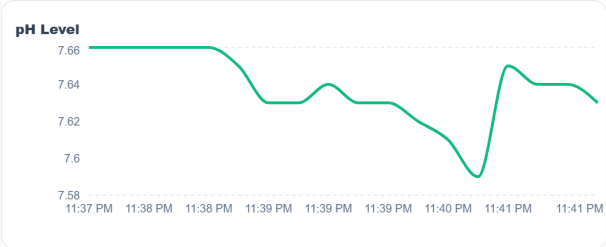
- CRITICAL DO: Turn ON main aerator immediately, increase surface agitation, and re-check DO within 10–15 minutes.
- TOXIC NH3: Perform partial water exchange, stop feeding for a short period, and ensure bio-filter is ON.
- HIGH TURBIDITY: Check inlet/outlet flow, reduce disturbance, and confirm filter/settling process.
- General guideline: Cross-check any critical reading with a manual test kit before major interventions.
- General guideline: Log actions taken (time, change, result) to improve future AI forecasting accuracy.

Based on latest samples (18 points). Forecast is dashed when enabled.

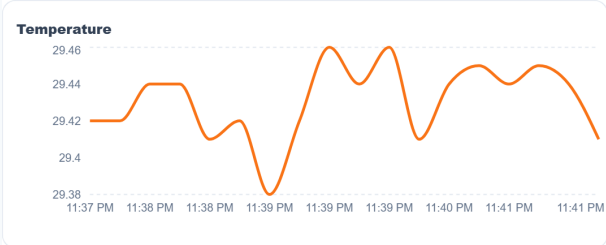
Dissolved Oxygen (mg/L)



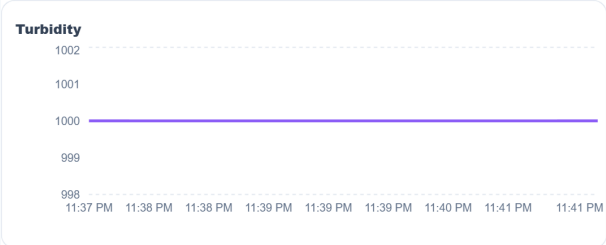
pH Level (pH)



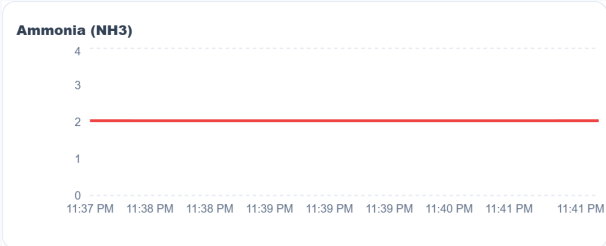
Temperature (°C)



Turbidity (NTU)



Ammonia (NH3) (mg/L)



Salinity (ppt)

